

CHAPTER-03

Professional Ethics for Engineers

Learning Points

1. What is Profession?
2. Is Engineering a Profession?
3. What is Ethics?
4. Differentiate between Ethics and Morality
5. What is Professional Ethics?
6. Types of Professional Ethics
7. What Is a Code of Ethics for Engineers?
8. What Are the Fundamental Principles of Engineering Code of Ethics?
9. Why Do Engineers Follow Code of Ethics?

3.1 What is a profession?

In order to gain clarity on the concept of a profession, let us look at its basic features which can be summarized as follows:

- (a) Profession is an occupation for which the individual has to acquire a special knowledge and skill.
- (b) The money they get for providing such a service is usually known as 'fee'.
- (c) Most of the professionals are regulated by a professional body, which frames the code of conduct to be followed by the member professionals. For example, Chartered Accountants in Bangladesh are regulated by a professional body known as Institutes of Chartered Accountants of Bangladesh, Doctors by Bangladesh Medical and Dental Council (BMDC), Engineers by the Institutions of Engineers, Bangladesh (IEB), Lawyers by Bangladesh Bar Council (BBC), Cricketers by International Cricket Council (ICC) and so on.

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- d) Professionals acquire the specialized knowledge mostly from colleges, universities or specialized institutes. In some cases, individuals also acquire such knowledge and skill through training or coaching by an expert in the same field, say for example, dancers and musicians, etc.
- e) Professionals usually work on their own and get a fee for their services and termed as those in practice. However, some of them may work in organizations as employees or consultants.
- (f) The primary objective of every profession is to provide service though they may charge a fee. They should not exploit the people using their knowledge or expertise.

3.2 Is Engineering a Profession?

Yes, engineering is widely considered a recognized profession as it satisfies all critical features of a profession discussed above. Those who are pursuing this profession are called engineers.

3.3 What is Ethics?

Ethics are a set of rules or standards established by the society, profession, or institutions to guide behavior in specific situation.

For example, reporting a potential safety hazard, treating colleagues with respect, adhering to a code of conduct, maintaining confidentiality, or being honest in research etc. are some of the examples of ethical behavior at workplace or in the professional settings.

3.4 Differentiate between Morality and Ethics

Morality and ethics, while related, represent distinct concepts.

1. Morality refers to an individual's personal beliefs about right and wrong, often shaped by culture and religion. Ethics, on the other hand, are a set of rules or standards established by the society, profession, or institutions to guide behavior in specific situation.
2. Morality comes from within, while ethics come from external sources such as societal norms, laws, and professional codes of conduct.
3. Morality may vary from person to person. While ethics may vary across different professions or organizations.
4. Morality is self-enforced, while ethics are often enforced by social, institutional, or professional pressure.
5. For example, A person's morality might lead him to always tell the truth, but the ethics of journalism might sometimes require withholding information for different reasons such as privacy concerns,, national security, or ongoing investigations. Similarly, a journalist might believe that it is morally wrong to publish a matter or fact that could harm an individual's privacy. However, as part of their professional ethics, they may be required to publish a personal matter or fact if it is newsworthy and in the public interest.

This illustrate how personal morals can sometimes conflict with the ethical standards of a profession.

3.5 What is Professional Ethics

Professional ethics are a set of moral principles that guide a person's conduct within a specific profession.

They dictate how a professional should interact with clients, colleagues, and the public, and are crucial for maintaining trust, integrity, and respect within the profession and society.

These principles often translate into codes of conduct established by professional organizations.

3.6 Types of Professional Ethics

Ethics may differ in every profession, some principle ethics are universal, e.g., being honest in the work and serving the people along with trustworthiness, respecting others, accountability, abiding by the rules and avoiding harming anyone etc.

Some particular professional bodies around the world determine the ethics of a specific profession. if anyone fails to do so, s/he becomes unworthy of being in that profession.

Professional ethics can be of many types based on the nature of professions. These may include the followings:

- a) Media and their Professional Ethics
- b) Judicial and Legal Professional Ethics
- c) Engineers' Professional Ethics
- d) Medical Professional Ethics
- e) Accountants' Professional Ethics
- f) Professional ethics for teachers and so on.

Let's discuss each of these professional ethics in brief:

a) Media and their Professional Ethics

Journalism ethics guide journalists in their work, ensuring accuracy, impartiality, truthfulness, and transparency in reporting the news. These principles are essential for maintaining public trust and upholding the integrity of the press.

In short, we can say that media ethics are about “seeking truth and reporting it.”

b) Judicial and Legal Professional Ethics

Legal professional ethics refers to the principles and standards that govern the conduct of lawyers. These principles ensure that lawyers act with integrity, honesty,, and competence, upholding the rule of law and maintaining public trust in the legal system.

Key aspects include avoiding conflicts of interest, maintaining client's confidentiality, and promoting access to legal service for all. They cannot force anyone to say something even if they know it's a lie, but they can make them confess the truth with tricks.

c) Engineer's Professional Ethics

Engineers' professional code outlines the expected ethical and professional conduct within the engineering field. These codes developed by professional engineering associations, guide engineers in fulfilling their responsibilities to the public, employers, clients and the profession itself.

They emphasize upholding the highest standards of integrity, competence, public safety and welfare, adhering to technical standards in their designs and constructions etc.

d) Medical Professional Ethics

Professional ethics for doctors revolve around prioritizing patient welfare and maintaining the integrity of the medical profession. A doctor shall protect human health. A doctor shall cure, alleviate, and console. A doctor shall help the ill to regain their health and the healthy to preserve theirs.

A doctor must refrain from doing unethical practices such as withholding or delaying treatment, taking kickbacks (gifts or commission) from pharmaceutical companies or diagnostic centers, overcharging for services etc.

e) Professional ethics for Accountants

Professional ethics for accountants consist of the following principles:

- a. Accountants must be honest and straightforward in performing all professional dealings.
- b. Accountants should avoid any bias or undue influence from others.
- c. Accountants must remain up-to-date with the latest accounting practices and regulations.
- d. Accountants must not disclose confidential information unless there is a legal or professional obligation to do so.

3.7 What Is a Code of Ethics for Engineers?

Professional ethics in engineering refers to the set of principles, values, and standards that guide the behavior and decision-making of engineers in their professional practice.

3.8 What Are the Fundamental Principles of Engineering Code of Ethics?

Every engineer in their line of duty is expected to:

- Uphold the safety, welfare and health of the society at all times
- Be truthful and honest when issuing public statements
- Offer their services in their areas of expertise only
- Be honest and trustful with their employers and clients
- Be honorable, responsible, lawful and ethical in their professional capacity to ensure good reputation and honor among the society

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- Treat all people with fairness with respect and dignity regardless of their personality, gender, race or other personal identification issues
- Use their knowledge and skills for the improvement of humanity
- Anticipate the current needs of society and fulfill them
- Every engineer has to follow and uphold these basic principles in their jobs without fail. In the event of a conflict or dilemma in fulfilling these basis codes, the engineer is bound to enforce the code based on the importance of each stakeholder.

3.9 Why Do Engineers Follow Code of Ethics?

1. Maintaining Safety

The work of engineers entails working on projects that impact the lives of members of society. Many times, their work involves building structures that people use daily. Therefore, engineers need to have a code of ethics that guide them in their daily work to ensure that they uphold the best interest of society.

Engineering code of ethics ensures that engineers put the safety of the members of the society first when doing their work.

2. Integrity and Honesty

Engineers, like all other professions, require integrity and honesty in their jobs. Codes of ethics enable them to be accountable for their actions. They act as guiding principles for determining what is right or wrong.

A code of ethics for engineers ensures that they remain honest in all their transactions. Even when faced with a dilemma or other pressures in their duty, when they adhere to the code of ethics, they will remain honest. By following the code of ethics, engineers can be frank with their clients, and at all times ensure that they keep the interests of the clients in mind.

3. Builds Public Trust and Confidence in the Profession

Engineering codes of ethics should be considered a crucial part of the engineering profession because it helps create faith among the population that engineers are ethical people who will do what is right even when no one else is looking. It also provides assurance to others that they have someone on their side with respect to safety or quality standards.

4. Ensures Ethical Decision Making in Areas of Uncertainty

Like all people, engineers face dilemmas in their line of duty. A code of ethics guides engineers in making these difficult decisions by ensuring that they choose what is moral. They give a clear guideline into what decision is ethical and serves the interests of the society rather than individual gain.

ETHICAL ANALYSIS

When confronted with a situation that seems to present ethical issues, how

should you analyze it? The following five-step process should help:

1. Identify and describe the facts clearly: Find out who did what to whom, and where, when, and how. In many instances, you will be surprised at the errors in the initially reported facts, and often you will find that simply getting the facts straight helps define the solution. It also helps to get the opposing parties involved in an ethical dilemma to agree on the facts.

ETHICAL ANALYSIS

2. Define the conflict or dilemma: Ethical, social, and political issues always reference higher values. The parties to a dispute all claim to be pursuing higher values (e.g., freedom, privacy, protection of property, and the free enterprise system). Typically, an ethical issue involves a dilemma: two diametrically opposed courses of action that support worthwhile values.

3. Identify the stakeholders: Every ethical, social, and political issue has stakeholders who have an interest in the outcome, who have invested in the situation, and usually who have vocal opinions. Find out of these groups and what they want. This will be useful later when designing a solution.

ETHICAL ANALYSIS

4. Identify the options that you can reasonably take. You may find that none of the options satisfy all the interests involved, but that some options do a better job than others. Sometimes arriving at a good or ethical solution may not always be a balancing of consequences to stakeholders.

5. Identify the potential consequences of your options. Some options may be ethically correct but disastrous from other points of view. Other options may work in one instance but not in other similar instances. Always ask yourself, “What if I choose this option consistently over time?”

CANDIDATE ETHICAL PRINCIPLES

Once your analysis is complete, what ethical principles or rules should you use to make a decision? What higher-order values should inform your judgment? Although you are the only one who can decide which among many ethical principles you will follow, and how you will prioritize them, it is helpful to consider some ethical principles with deep roots.

1. Putting yourself into the place of others, and thinking of yourself as the object of the decision, can help you think about fairness in decision making.
2. If an action is not right for everyone to take, it is not right for anyone.

CANDIDATE ETHICAL PRINCIPLES

3. If an action cannot be taken repeatedly, it is not right to take at all (Descartes' rule of change). This is the slippery-slope rule: An action may bring about a small change now that is acceptable, but if it is repeated, it would bring unacceptable changes in the long run.
4. Take the action that achieves the higher or greater value (Utilitarian Principle). This rule assumes you can prioritize values in a rank order and understand the consequences of various courses of action.

CANDIDATE ETHICAL PRINCIPLES

5. Take the action that produces the least harm or the least potential cost (Risk Aversion Principle). Some actions have extremely high failure costs of very low probability (e.g., building a nuclear generating facility in an urban area) or extremely high failure costs of moderate probability (speeding and automobile accidents). Avoid these high-failure-cost actions, paying greater attention to high-failure-cost potential of moderate to high probability.

6. Assume that virtually all tangible and intangible objects are owned by someone else unless there is a specific declaration otherwise. (This is the ethical “no free lunch” rule.) If something someone else has created is useful to you, it has value, and you should assume the creator wants compensation for this work.